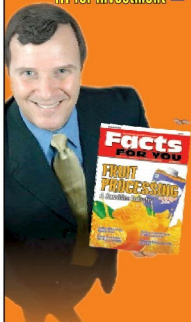


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years of battery life in low-duty-cycle applications.

Power management. It is critical for medical devices to consume minimal power, especially if these are intended to be worn on or implanted inside the human body. The power consumption of medical devices can affect the implant's lifetime, the amount of heat generated by the device, battery weight and battery cost.

Microcontrollers and other ICs for medical applications achieve low power consumption in a number of ways. Multiple low-power modes, for instance, give designers the flexibility to turn some or all of the MCU functions on or off.

The MCU must support low power in active mode, standby mode and reduced time in an active state. The portable device, and thus the MCU, will be in 'off' or lower power state most of the time. However, the device also needs to maintain some sort of function such as clock/calendar or alarm even when not in use. Thus minimising its awake time is the key to extending the battery life. The MCU design should be able to wake the clock and analogue circuits for fast measurements and then allow the MCU to settle back to a low-power state.

A range of ultra-low-power microcontrollers (STM32L and STM8L series from STMicroelectronics) based on a proprietary 130nm ultra-low-leakage process technology are available. These minimise overall energy consumption without compromising on speed and functionality.

In parallel with the drive towards lower current draw, the supply rails required by medical devices are also coming down. While this is relatively easy to achieve in digital circuits as the industry follows the Moore's Law curve, it is tough for mixed-signal measurement systems. Problems with leakage and signal-to-noise ratio abound. These problems, however, are gradually being brought under control, thus easing the design of the overall



Onyx-II (Model 9560) fingertip pulse oximeter

power management system.

Connectivity. It is becoming more common for portable medical-electronic devices to communicate with each other or with other devices, either point-to-point or over a network. Connected devices create a number of new capabilities. The simplest may be the one where a device is able to transfer data to a personal computer for storage or analysis. This has so far been done via serial bus or infrared, but latest devices use USB.

There are also systems that use multiple sensors on a patient to monitor temperature, blood pressure, pulse rate, blood oxygen level, etc. In a networked system, all of these sensors can provide data to a central monitor. Gaining in popularity are systems that gather data from one or more sensors and transfer it to a server via the Internet. This allows patients, or their doctors, to monitor their conditions from anywhere.

Wireless connectivity is redefining the way medicine is practiced. A wireless remote monitoring device can take many forms, but most consist of a wireless sensor connected to a consumer medical device like an ECG, heart monitor or glucometer. These